Data Preprocessing Project : Sentiment Analysis

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1 Introduction

n the world of online education, Massive Open Online Courses (MOOCs) have become super popular. They offer a variety of courses and allow people from all over the world to learn at their own pace. One key way we understand how well these courses are working is through the feedback from learners, like reviews and comments. But here's the hitch: going through all these comments manually is a real tough job. This is where the idea of sentiment analysis for MOOC learner reviews comes in.

2 Define the Problem

2.1 Problem and context Statement

The big issue here is that going through all the learner reviews one by one takes a lot of time and effort. As more and more people join these courses, it's becoming a bit of a headache. It's not just about reading what people say it's about understanding if they're happy, if there are things that need fixing, and what they really like. Doing this by hand slows things down, and there's a risk of missing important stuff. So, the challenge is to figure out a way to do this sentiment analysis basically, understanding if the feedback is positive, negative, or neutral without spending forever on it. Provide the context of the problem and its significance. Think of it like this: there's a huge pile of comments, and each one is like a little piece of gold with important information. But digging through this pile manually is like looking for treasure in the dark without a map. We need a smarter way to use technology to help us make sense of all these comments. The tricky part is that these comments are not just numbers; they are words, and lots of them. So, we need a clever computer program that understands the language people use and can quickly tell us if they're happy, unhappy, or somewhere in between.

2.2 Problem in General

so how can we leverage advanced machine learning and natural language processing techniques to automate sentiment analysis of MOOC learner reviews, effectively mining the extensive textual data to enhance course quality, identify areas for improvement, and gain actionable insights into learner satisfaction and preferences?

3 Data Collection

he dataset used in this project was collected from various Massive Open Online Courses (MOOCs) and comprises over 9 million comments. These comments are contributed by diverse individuals engaging with different courses, reflecting a wide range of perspectives and languages. The dataset includes feedback

on course content, experiences, and preferences, making it a rich source for analyzing learner sentiments and extracting valuable insights.

The comments cover a variety of topics and are expressed in different languages, adding a layer of complexity and diversity to the dataset. This wealth of data offers the opportunity to explore patterns, sentiments, and preferences among MOOC learners, ultimately contributing to a comprehensive understanding of the effectiveness and appeal of online courses.

Out[45]:

	id	course_id	rate	date	display_name	comment	comment_length	review
0	88962892	3173036	1.0	2021-06-29 18:54:25-07:00	Rahul	I think a beginner needs more than you think.\...	148	Negative
1	125535470	4913148	5.0	2022-10-07 11:17:41-07:00	Marlo	Aviva is such a natural teacher and healer/hea...	250	Positive
2	68767147	3178386	3.5	2020-10-19 06:35:37-07:00	Yamila Andrea	Muy buena la introducción para entender la bas...	364	Positive
3	125029758	3175814	5.0	2022-09-30 21:13:49-07:00	Jacqueline	This course is the best on Udemy. This breakd...	379	Positive
4	76584052	3174896	4.5	2021-01-30 08:45:11-08:00	Anthony	I found this course very helpful. It was full ...	130	Positive
5	124129784	4693438	1.0	2022-09-20 11:30:29-07:00	Jiang	nothing information there,	26	Negative
6	121769970	4693272	3.5	2022-08-22 11:48:56-07:00	Kenneth	Multiple spelling mistakes and one or more que...	70	Positive
7	57260120	3168632	5.0	2020-06-02 22:56:44-07:00	Tony	Very unique way of teaching. Simple but Powerf...	72	Positive
8	77427106	3188362	4.0	2021-02-09 21:19:29-08:00	HIROKO	グルテンフリーのポイントがよくわかり生活習慣を見直すきっかけになりました。ありがとうございました。	49	Positive
9	103846020	4164550	4.5	2022-01-02 09:04:53-08:00	Jess	Good Course! Information is well organized cl...	70	Positive

Activate Windows
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Figure 1: Data Frame

4 Exploratory Data Analysis (EDA)

4.1 Basic Statistics

We can start performing statistical analysis using various pandas functions. Here are some common statistical operations we can perform:

• **Descriptive Statistics:** We use the describe() method to get a summary of basic statistical measures for each numerical column in our dataset.

```
In [29]: df.columns
Out[29]: Index(['id', 'course_id', 'rate', 'date', 'display_name', 'comment'], dtype='object')

In [ ]:
```

Figure 2: Columns of data frame.

```
In [42]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9411727 entries, 0 to 9411726
Data columns (total 7 columns):
 #   Column          Dtype
---  ------          -----
 0   id              int64
 1   course_id       int64
 2   rate            float64
 3   date            object
 4   display_name    object
 5   comment         object
 6   comment_length  int64
dtypes: float64(1), int64(3), object(3)
memory usage: 502.6+ MB
```

Figure 3: Informations on Data Frame

In [4]: df.describe()

Out[4]:

	id	course_id	rate
count	9.411727e+06	9.411727e+06	9.411727e+06
mean	5.625454e+07	1.589933e+06	4.508870e+00
std	3.754655e+07	1.100059e+06	8.587239e-01
min	5.400000e+01	1.769000e+03	5.000000e-01
25%	2.256843e+07	7.641640e+05	4.000000e+00
50%	5.296542e+07	1.333640e+06	5.000000e+00
75%	8.625520e+07	2.243978e+06	5.000000e+00
max	1.267091e+08	4.913148e+06	5.000000e+00

Figure 4: Describe Data Frame

• **Column-wise Statistics:**

We can apply this only if our calumn are numerical , Basically we when wanted to calculate the standard deviation we had errors of 'could not convert string to float: 'Rahul'.

• **Counting and Frequency:**

Counting the number of occurrences of unique values in some columns.

```
course_id_percentages = df['course_id'].value_counts(normalize=True)

In [10]: course_id_counts

Out[10]: course_id
         567828     39040
         793796     36101
         625204     31001
         1565838    27723
         914296     27540
                    ...
         327548         1
         4226992        1
         4175632        1
         2389532        1
         3173036        1
         Name: count, Length: 162995, dtype: int64

In [11]: course_id_percentages

Out[11]: course_id
         567828     4.148017e-03
         793796     3.835747e-03
         625204     3.293869e-03
         1565838    2.945581e-03
         914296     2.926137e-03
                        ...
         327548     1.062504e-07
         4226992    1.062504e-07
         4175632    1.062504e-07
         2389532    1.062504e-07
         3173036    1.062504e-07
```

Figure 5: CourseID Counts

```
In [10]: course_id_counts

Out[10]: course_id
         567828     39040
         793796     36101
         625204     31001
         1565838    27723
         914296     27540
                    ...
         327548         1
         4226992        1
         4175632        1
         2389532        1
         3173036        1
         Name: count, Length: 162995, dtype: int64
```

Figure 6: Number Of courses

- **Checking Missing Values:**

```
In [12]: df.isnull().sum()

Out[12]: id                 0
         course_id          0
         rate               0
         date               0
         display_name    75387
         comment          6548
         dtype: int64
```

Figure 7: Missing Values In Each Feature

```
In [38]: print(round(df.isnull().mean() * 100, 2))

         id              0.00
         course_id       0.00
         rate            0.00
         date            0.00
         display_name    0.80
         comment         0.07
         comment_length  0.00
         dtype: float64

In [ ]:
```

Figure 8: Percenatge Of Missing Value

- **Correlation:**

Calculate the correlation matrix to understand the relationships between numerical variables.

```
In [23]: numerical_columns = ['id', 'course_id', 'rate']
         df_numerical = df[numerical_columns]
         correlation_matrix = df_numerical.corr()

In [27]: correlation_matrix
```

Out[27]:

	id	course_id	rate
id	1.000000	0.568787	0.009862
course_id	0.568787	1.000000	0.019179
rate	0.009862	0.019179	1.000000

Figure 9: Corr()

• **Unique Values:**
List of distinct categories in each column (For Categorical Data)

```
    print("\n")

Unique values in rate:
rate
5.0    5899806
4.0    1165812
4.5    1106120
3.5     390865
3.0     379772
1.0     174177
2.5     130326
2.0     110983
1.5      40134
0.5      13713
4.9          3
4.4          3
3.7          2
4.6          2
1.2          2
4.1          2
4.8          2
4.2          2
3.9          1
Name: count, dtype: int64
```

Figure 10: Unique Values of Rate

```
Unique values in display_name:
display_name
Anonymized              147793
David                    50781
Michael                  44334
Daniel                   39054
John                     36334
                          ...
Souregl                      1
Reisson Luis Saavedra        1
khuram                       1
1                            1
Rao Saurabh                  1
Name: count, Length: 1420901, dtype: int64
```

Figure 11: Unique Values of names

4.2 Visualization

isualization refers to the creation and interpretation of graphical representations of data. Visualization is a crucial component of EDA because it allows analysts to gain insights into the patterns, trends, and relationships within a dataset more effectively than through raw data alone. Visualizations provide a way to explore and communicate complex information in a more accessible and intuitive

manner.

- **Box Plots:**
  Visualize the distribution of numerical variables using box plots.

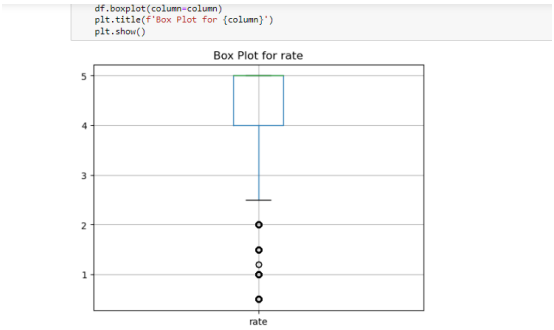


Figure 12: Box Plots

- **Histograms:**
  Visualize the distribution of a numerical variable (ID , Course ID ,Rate) using histograms.

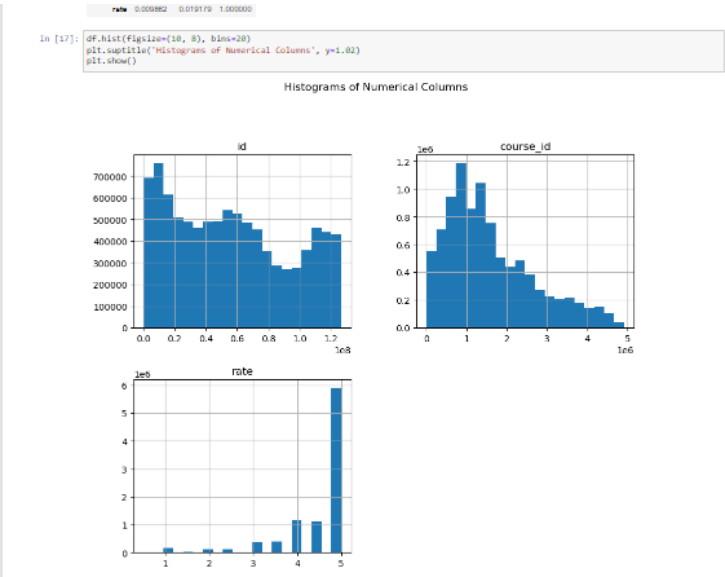


Figure 13: Histograms

- **Visualize the Distribution of Rate:** A Distribution of Course Ratings

is a bar chart will be used to visualize the distribution of course ratings, providing insights into the prevalence of each rating. This will help understand the overall sentiment of the reviews.

```
plt.title('Distribution of Rates')
plt.show()
```

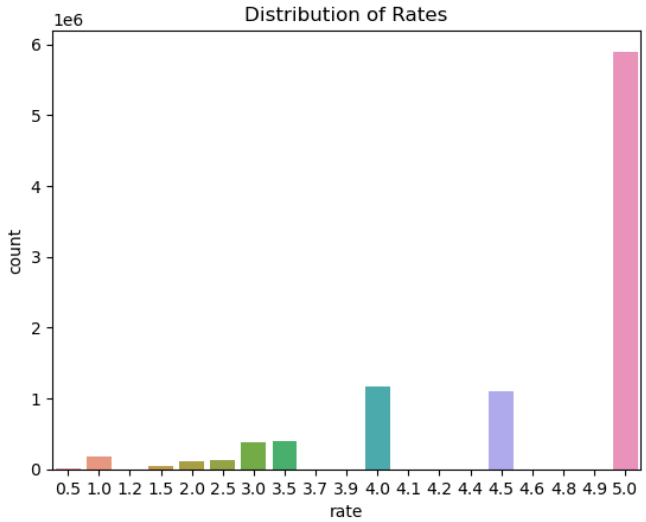


Figure 14: Distribution of rates

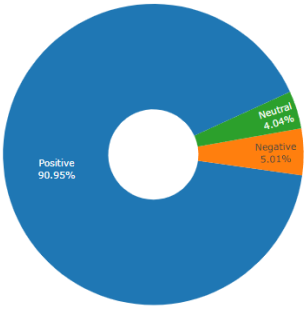
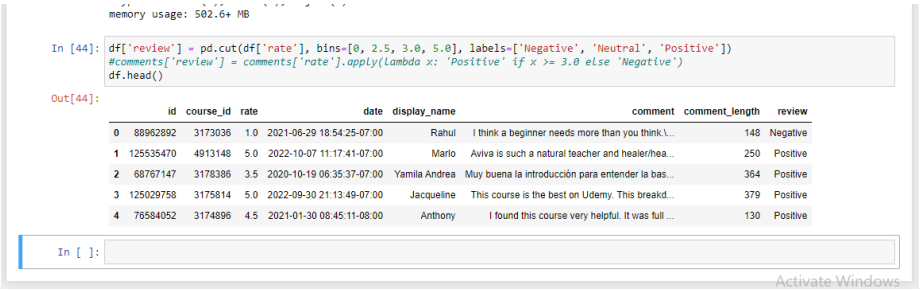


Figure 15: Distribution of rates

• **Review (Positive, Negative, Neutral):**

Another column, 'Review,' has been added based on ratings. Comments with a rating greater than 3 are labeled as 'Positive,' less than 3 as 'Negative,' and equal to 3 as 'Neutral', which would absolutely affording us the way to create our model in the future coming steps.



memory usage: 502.6+ MB

In [44]:
```
df['review'] = pd.cut(df['rate'], bins=[0, 2.5, 3.0, 5.0], labels=['Negative', 'Neutral', 'Positive'])
#comments['review'] = comments['rate'].apply(lambda x: 'Positive' if x >= 3.0 else 'Negative')
df.head()
```

Out[44]:

	id	course_id	rate	date	display_name	comment	comment_length	review
0	88962892	3173036	1.0	2021-06-29 18:54:25-07:00	Rahul	I think a beginner needs more than you think.\...	148	Negative
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4	76584052	3174896	4.5	2021-01-30 08:45:11-08:00	Anthony	I found this course very helpful. It was full ...	130	Positive

In [ ]:

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Figure 16: New Feature 'Review'

• **The Distribution of Comment Lengths in The Dataset:**

This visualization provides insights into the range and frequency of comment lengths also :

- Identifying common comment lengths can inform strategies for content presentation or engagement.
- Analyzing outliers may reveal particularly lengthy or short comments that might warrant specific attention.

```
plt.xlabel('Length of Comments')
plt.ylabel('Frequency')
plt.show()
```

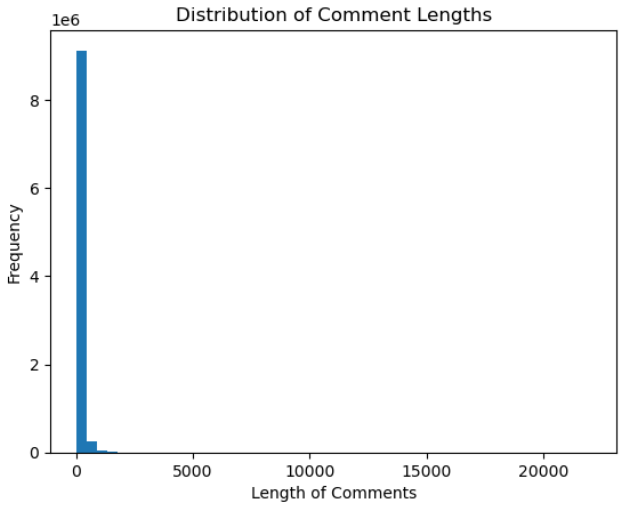


Figure 17: Distribution of Comments Length

• **Visualization of Relationship Between Rating and Comment Length:** his analysis aims to understand whether there's a correlation between the sentiment of reviews (as indicated by the rating) and the length of comments

This visualization will help uncover any potential correlations or patterns between course ratings and the length of comments , and also would help observing patterns may provide insights into the relationship between sentiment and user engagement , In addition to that Understanding how comment length varies with different ratings can inform content creators about user behavior.

Their actionable Insights are understanding the sentiment distribution can guide strategies for improving courses with lower ratings. Analyzing comment length in relation to sentiment can inform content creators about the nature of reviews, guiding improvements in user engagement or content delivery.

```
In [34]: df['comment_length'] = df['comment'].fillna('').apply(len)
         sns.boxplot(x='rate', y='comment_length', data=df)
         plt.title('Relationship Between Rate and Comment Length')
         plt.show()
```

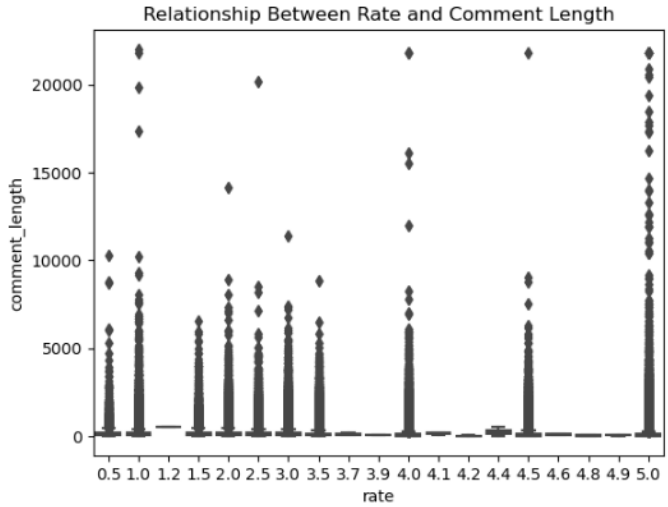


Figure 18: Relationship Between Rate And Comment Length

```
courses_per_year.plot(kind='bar', color='lightgreen')
plt.title('Number of Added Courses per Year')
plt.xlabel('Year')
plt.ylabel('Number of Courses')

plt.tight_layout()
plt.show()
```

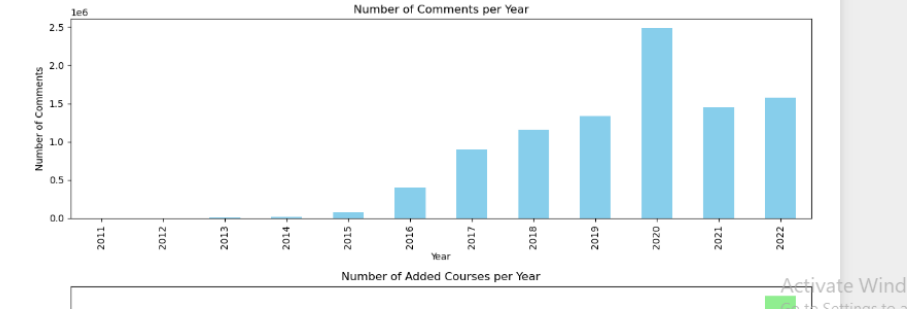


Figure 19: Comments Per Year